

Credit Risk Assessment & Loan Default Analysis

Executive Credit Risk Assessment Report

SQL | Power BI | Business Intelligence | Credit Risk Analytics | Executive Reporting

Portfolio Project

Prepared by: Dayo Rex Afariogun

Role: Senior Business Analyst | Financial Data Analytics & Business Intelligence

About This Project

This project demonstrates how modern data analytics can support evidence-based lending decisions by transforming loan application data into meaningful credit risk intelligence. Using **SQL** for data extraction and **Power BI** for data transformation, visualisation, and executive reporting, the analysis evaluates applicant characteristics that influence loan approval decisions and overall portfolio quality.

Rather than simply reporting approval and rejection statistics, this project applies business intelligence techniques to assess borrower risk, identify lending patterns, evaluate portfolio concentration, and explain how analytical insights can improve profitability while reducing exposure to loan defaults.

The project integrates technical analytics with financial reasoning by connecting lending decisions to broader business outcomes, including profitability, liquidity, regulatory compliance, capital preservation, and shareholder value.

The resulting executive dashboard enables decision-makers to monitor key lending indicators, identify emerging credit risks, evaluate borrower profiles, and strengthen lending strategies through data-driven decision-making.

Executive Summary

Credit risk management remains one of the most important responsibilities of every financial institution. Every lending decision carries a degree of uncertainty because borrowers may fail to meet their repayment obligations. Effective credit assessment therefore requires balancing commercial growth with prudent risk management to maximise profitability while protecting the organisation from excessive loan losses.

This project analyses a dataset containing **5,000 loan applications** using **Microsoft SQL Server** for data extraction and **Power BI** for data transformation, modelling, dashboard development, and interactive visualisation. The objective was to identify the key factors influencing loan approval decisions, evaluate borrower risk profiles, and develop an executive-level dashboard capable of supporting strategic lending decisions.

The analysis reveals that **1,151 loan applications (23%) were approved**, while **3,849 applications (77%) were rejected**, indicating that the institution follows a highly selective lending policy designed to minimise default risk and maintain a high-quality loan portfolio.

Beyond measuring approval rates, the project explores how borrower characteristics—including employment status, credit score, income, years of work experience, age, gender, education, and geographic location—influence lending outcomes. These insights enable management to identify low-risk borrowers, monitor portfolio quality, optimise lending strategies, and improve long-term financial performance.

From a business perspective, the project demonstrates how credit analytics contributes to improved decision-making by reducing bad debts, protecting capital, improving cash flow, strengthening regulatory compliance, and supporting sustainable lending growth.

From a professional perspective, this project showcases practical competencies in SQL querying, data modelling, Power BI dashboard development, DAX calculations, business intelligence, financial analysis, executive reporting, and data storytelling—capabilities highly valued across banking, financial services, insurance, fintech, consulting, and multinational organisations.

Business Context

Financial institutions operate within an environment where lending decisions directly influence profitability, liquidity, regulatory compliance, and long-term organisational sustainability.

While lending generates interest income and drives business growth, poor credit decisions expose institutions to loan defaults, increased impairment provisions, capital erosion, and regulatory scrutiny. As loan portfolios expand, manually assessing borrower risk becomes increasingly inefficient and inconsistent, making data-driven credit assessment essential for modern financial institutions.

Every loan application contains valuable information regarding an applicant's financial capacity, employment stability, creditworthiness, and repayment potential. By analysing these characteristics systematically, financial institutions can improve lending decisions while balancing profitability with acceptable levels of credit risk.

Business Intelligence provides a structured framework for transforming large volumes of applicant data into actionable insights capable of supporting both operational and strategic decision-making.

This project addresses these challenges by developing an executive credit risk dashboard that analyses loan applications across multiple borrower dimensions, including:

- Loan approval performance
- Credit score distribution
- Employment status
- Income characteristics
- Work experience
- Age demographics
- Gender distribution
- Geographic loan demand
- Portfolio risk indicators

The dashboard enables executives to monitor lending performance, evaluate portfolio quality, identify high-risk borrower profiles, and strengthen future lending strategies using objective, evidence-based analysis.

Beyond technical reporting, the project illustrates how data analytics can support responsible lending while balancing commercial growth, financial stability, customer service, and regulatory expectations.

Strategic Project Objectives

The primary objective of this project was to develop an executive-level credit risk assessment solution capable of improving lending decisions through advanced data analytics and business intelligence.

Specifically, the project aimed to:

- Analyse historical loan application data.
- Measure overall loan approval and rejection performance.
- Identify the primary factors influencing lending decisions.
- Evaluate borrower characteristics associated with lower default risk.
- Assess the relationship between employment status and loan approval.
- Analyse the influence of credit scores on lending outcomes.
- Evaluate borrower demographics and geographic loan demand.
- Develop an interactive Power BI dashboard for executive reporting.
- Translate analytical findings into practical business recommendations.
- Explain how lending decisions influence financial performance and portfolio quality.
- Demonstrate the practical application of SQL and Power BI within credit risk management.

Collectively, these objectives provide management with a comprehensive framework for evaluating lending performance while supporting responsible, data-driven credit decision-making.

Executive Business Questions

The analysis was designed to answer several strategic questions commonly faced by banks, lending institutions, fintech companies, and credit risk teams.

Loan Portfolio Performance

- How many loan applications were received?
- What percentage of applications were approved?
- What percentage were rejected?
- What does the approval rate indicate about the organisation's lending strategy?

Credit Risk Assessment

- Which borrower characteristics are associated with lower default risk?
- Which customer profiles present higher lending risk?
- How effectively does the current approval process minimise future loan losses?

Employment Analysis

- Does employment status significantly influence loan approval decisions?
- Which employment categories receive the highest approval rates?
- Does employment stability improve repayment probability?

Credit Score Analysis

- How strongly does credit score influence approval decisions?
- Which credit score ranges demonstrate the lowest credit risk?
- Should lending policies place greater emphasis on credit history?

Income & Work Experience

- Does higher income improve approval probability?
- How does employment experience affect borrower risk?
- Are experienced borrowers consistently lower-risk customers?

Demographic Analysis

- Does age influence approval decisions?
- Is gender associated with lending outcomes?
- Are lending decisions primarily driven by financial indicators rather than demographic characteristics?

Geographic Analysis

- Which cities generate the highest loan demand?
- Does geographic diversification reduce portfolio concentration risk?
- Are lending opportunities evenly distributed across markets?

Executive Decision Support

- What are the primary drivers of successful loan approvals?
- Which risk factors require continued monitoring?
- What strategies can strengthen portfolio quality while supporting sustainable lending growth?

Answering these questions enables executives to move beyond descriptive reporting and adopt a proactive approach to credit risk management.

Technology Stack

Technology	Business Purpose
Microsoft SQL Server	Storage and management of loan application data
SQL	Data extraction, filtering, joins, aggregation, and querying
Power BI	Interactive dashboard development and executive reporting
Power Query	Data cleaning, transformation, and preparation
DAX	KPI calculations and business metrics
Microsoft Excel	Initial data validation and quality assurance
Business Intelligence Dashboard	Executive reporting and lending decision support

Project Methodology

This project followed a structured Business Intelligence methodology that transformed raw loan application data into actionable credit risk insights for executive decision-making.

Phase 1 — Business Understanding

The project began by identifying the institution's primary lending objectives and understanding the key business questions surrounding loan approvals, portfolio quality, borrower risk, and credit performance.

This ensured that the analysis remained aligned with strategic lending objectives rather than focusing solely on descriptive statistics.

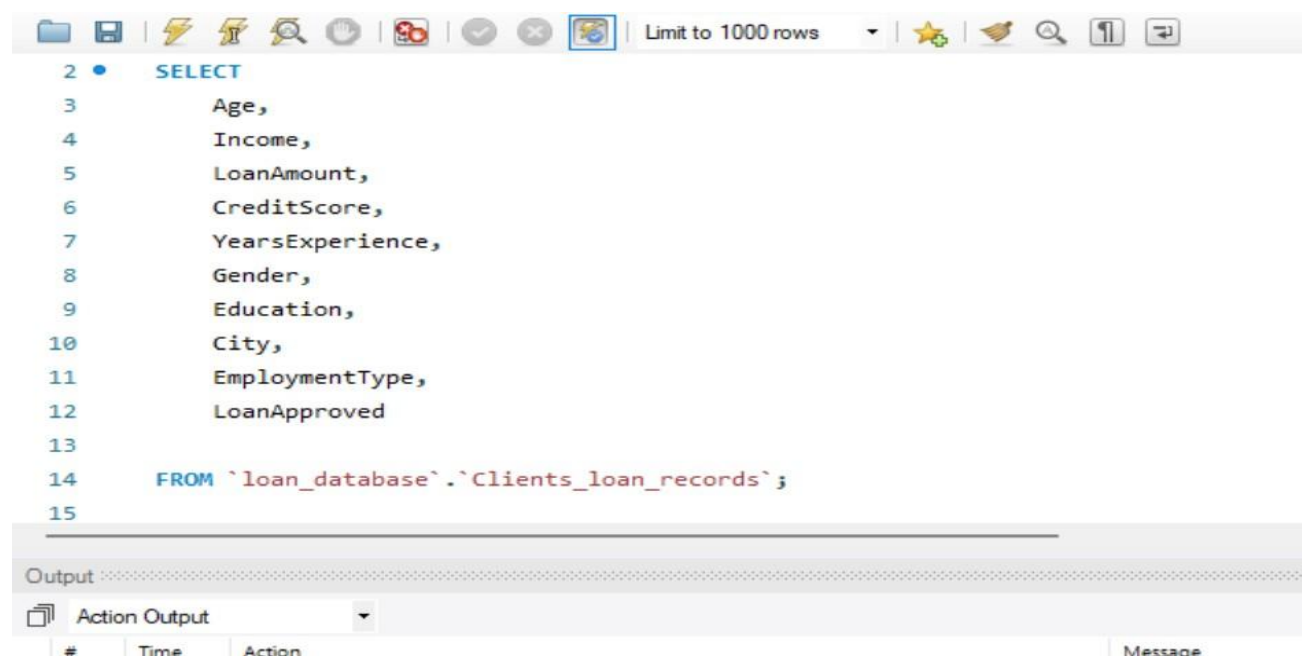
Phase 2 — Data Extraction

Loan application data was extracted from the relational database using Microsoft SQL Server.

SQL queries were developed to retrieve applicant information including:

- Age
- Income
- Loan Amount
- Credit Score
- Years of Experience
- Employment Type
- Education
- Gender
- City
- Loan Approval Status

SQL provided efficient extraction of clean, structured data suitable for downstream modelling and analysis.



Phase 3 — Data Preparation

Following extraction, the dataset was imported into Power BI where Power Query was used to prepare the data for analysis.

Preparation activities included:

- Data validation
- Data cleaning
- Handling missing values
- Standardising data formats
- Removing duplicate records
- Creating calculated fields
- Preparing analytical dimensions

Establishing a high-quality dataset ensured that subsequent analytical outputs accurately reflected borrower characteristics and lending performance.

Phase 4 — Data Analysis

Once prepared, the data was analysed using Power BI and DAX measures.

Multiple analytical dimensions were explored, including:

- Loan approval rates
- Credit score analysis
- Employment status
- Income characteristics
- Work experience
- Demographic trends
- Geographic distribution
- Portfolio risk indicators

This phase transformed raw loan application records into meaningful business intelligence capable of supporting executive lending decisions.

Phase 5 — Dashboard Development

Interactive Power BI dashboards were designed to provide executives with a comprehensive view of lending performance.

The dashboard integrates KPI cards, interactive filters, charts, and borrower segmentation to enable dynamic exploration of portfolio performance across multiple dimensions.

This interactive design allows management to quickly identify lending trends, monitor risk exposure, and evaluate portfolio quality.

Phase 6 — Business Interpretation

Analytical outputs were interpreted from both commercial and financial perspectives.

Rather than presenting approval statistics alone, the analysis explains:

- Why certain borrower groups receive approvals
- How borrower characteristics influence default risk
- What risks exist within the current lending strategy
- How portfolio quality affects financial performance
- Which opportunities exist for improving lending effectiveness

This transforms descriptive analytics into strategic business intelligence.

Phase 7 — Strategic Recommendations

Finally, analytical findings were translated into practical recommendations designed to strengthen lending decisions while maintaining sustainable portfolio growth.

Recommendations focus on:

- Credit risk reduction
- Portfolio quality improvement
- Lending policy optimisation
- Regulatory compliance
- Predictive risk modelling
- Customer assessment
- Long-term financial sustainability

By combining technical analytics with financial expertise and business interpretation, this project demonstrates how modern Business Intelligence supports effective credit risk management and evidence-based lending decisions.

Dashboard Overview

The Credit Risk Assessment & Loan Default Analysis dashboard provides executives with a comprehensive view of lending performance across the entire loan approval lifecycle. Rather than focusing solely on loan approvals and rejections, the dashboard enables decision-makers to understand the underlying borrower characteristics that influence lending outcomes and overall portfolio quality.

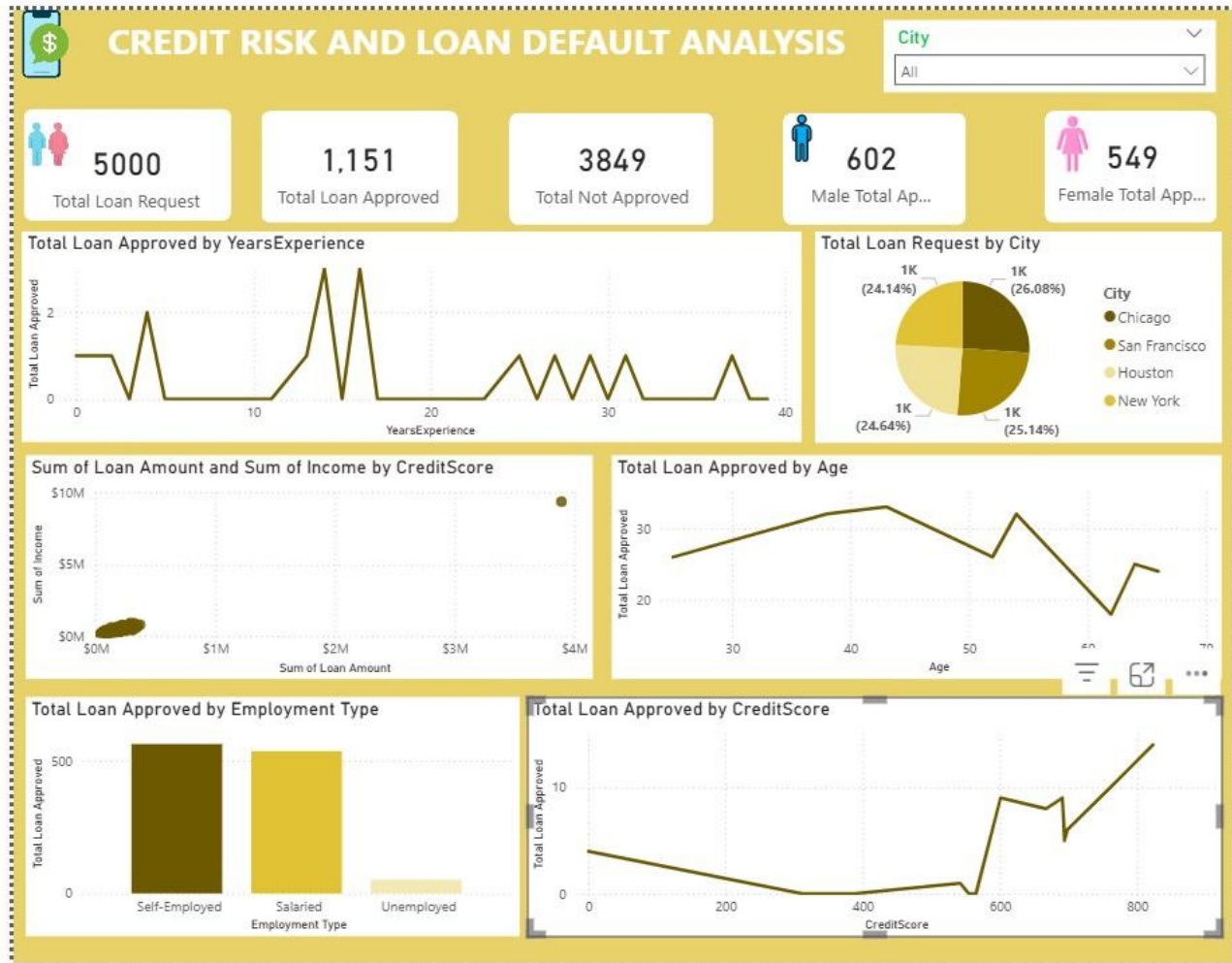
Designed using **Power BI**, the dashboard consolidates multiple dimensions of loan performance into a single interactive reporting environment, allowing users to evaluate applicant demographics, employment status, creditworthiness, geographic distribution, and loan approval trends.

The dashboard transforms raw lending data into meaningful business intelligence, enabling executives to identify risk concentrations, monitor lending policies, evaluate portfolio quality, and make informed strategic lending decisions.

The dashboard contains interactive visualisations covering:

- Overall Loan Portfolio Performance
- Loan Approval Status
- Credit Score Distribution

- Employment Status
- Annual Income Analysis
- Years of Work Experience
- Applicant Age Distribution
- Gender Distribution
- Educational Qualification
- Geographic Loan Distribution
- Interactive Loan Filters



Each visual contributes to a broader understanding of borrower behaviour and lending performance, allowing management to move beyond descriptive reporting towards proactive credit risk management.

Unlike traditional operational reports, this dashboard provides executives with a real-time analytical view of lending operations, supporting faster, evidence-based decision-making while strengthening portfolio quality.

KPI Analysis

Key Performance Indicators (KPIs) provide an executive summary of the institution's lending performance by presenting critical lending metrics in an easily interpretable format.

These indicators enable management to monitor lending effectiveness, assess portfolio quality, and evaluate whether current lending strategies are aligned with organisational risk appetite.

The dashboard summarises the following key metrics.

KPI	Business Interpretation
-----	-----
Total Loan Applications	Overall lending demand received by the institution
Approved Loans	Number of successfully approved applications
Rejected Loans	Number of declined applications
Loan Approval Rate	Percentage of applications approved
Loan Rejection Rate	Percentage of applications rejected

Total Loan Applications

The institution processed **5,000 loan applications**, providing a sufficiently large dataset for meaningful credit risk analysis and lending performance evaluation.

A high application volume demonstrates healthy market demand for lending products and provides management with opportunities to build a diversified loan portfolio.

However, increasing application volume also requires stronger analytical capabilities to ensure lending decisions remain accurate, consistent, and compliant with internal credit policies.

From a business perspective, loan applications represent the primary source of future revenue generation because approved facilities generate interest income throughout the repayment period.

Approved Loans

The dashboard indicates that **1,151 loan applications were approved**, representing approximately **23% of all applications received**.

The relatively conservative approval rate suggests that the institution prioritises portfolio quality over aggressive loan growth.

This lending approach reduces exposure to potential loan defaults by approving only applicants who satisfy established credit assessment criteria.

Although lower approval rates may reduce short-term lending revenue, they often strengthen long-term financial performance by reducing future credit losses.

Rejected Loans

A total of **3,849 applications were rejected**, representing approximately **77% of all applications submitted**.

The high rejection rate indicates that the institution maintains rigorous underwriting standards designed to preserve asset quality and minimise non-performing loans.

Rather than interpreting rejections negatively, management should view them as evidence of disciplined credit governance that protects shareholders' capital and strengthens financial resilience.

Nevertheless, continuous review of rejected applications is essential to ensure credit policies remain commercially competitive without compromising acceptable risk levels.

Loan Approval Rate

The overall approval rate of approximately **23%** reflects a highly selective lending strategy.

A conservative approval policy generally indicates that loan decisions are heavily influenced by objective borrower characteristics such as:

- Credit history
- Employment stability
- Income capacity
- Financial behaviour
- Repayment capability

While this strategy improves portfolio quality, management should continuously evaluate whether approval thresholds appropriately balance commercial growth with prudent risk management.

Overly restrictive lending policies may result in missed revenue opportunities, whereas excessively relaxed policies may increase future impairment losses.

Loan Rejection Rate

The rejection rate of approximately **77%** demonstrates the institution's strong focus on credit risk mitigation.

Although rejecting a large proportion of applicants reduces immediate loan disbursement volumes, it also helps preserve portfolio quality by limiting exposure to borrowers with elevated default risk.

The institution should periodically reassess rejection criteria to ensure lending policies remain aligned with changing market conditions, economic trends, and customer credit behaviour.

Executive Insights

Loan Approval Analysis

Loan approval analysis provides valuable insight into the overall effectiveness of the institution's lending strategy.

The dashboard demonstrates that lending decisions are not distributed randomly but instead reflect structured credit assessment practices based on measurable borrower characteristics.

The relatively low approval rate suggests that the institution prioritises:

- Portfolio quality

- Sustainable profitability
- Regulatory compliance
- Capital preservation
- Credit risk reduction

This disciplined lending approach reduces future loan impairment while maintaining a healthier loan portfolio.

From a strategic perspective, management should continuously monitor approval trends to identify changing borrower behaviour, emerging credit risks, and opportunities for controlled portfolio expansion.

Credit Score Analysis

Credit score remains one of the strongest predictors of borrower repayment behaviour and future credit performance.

The dashboard demonstrates a clear relationship between creditworthiness and lending outcomes.

Applicants with stronger credit profiles are significantly more likely to receive loan approvals because they present lower expected default risk.

Conversely, applicants with weaker credit histories require additional assessment due to their increased probability of repayment difficulties.

From a business perspective, incorporating credit score into lending decisions supports:

- Reduced default risk
- Improved portfolio quality
- Lower impairment provisions
- More accurate borrower segmentation
- Sustainable interest income generation

However, credit score should not operate as the sole approval criterion.

Modern credit risk assessment benefits from combining credit history with other borrower characteristics such as employment stability, income capacity, debt obligations, and repayment behaviour.

This multi-dimensional approach produces more balanced lending decisions while improving predictive accuracy.

Employment Status Analysis

Employment stability represents one of the most significant indicators of future repayment capacity.

The dashboard evaluates applicant employment categories to determine how employment status influences lending outcomes.

Borrowers with stable employment generally demonstrate:

- Consistent income streams
- Improved repayment capacity
- Lower financial uncertainty
- Reduced default probability

Consequently, employed applicants tend to experience higher approval rates than applicants with unstable or irregular income sources.

From a commercial perspective, employment analysis enables lending institutions to better understand customer risk profiles while improving underwriting decisions.

Management can also use these insights to develop specialised lending products tailored to different employment groups, balancing commercial growth with acceptable risk exposure.

Employment analysis therefore supports both customer acquisition and prudent credit risk management.

Income and Financial Capacity

Although employment provides evidence of financial stability, borrower income determines actual repayment affordability.

Higher-income applicants generally possess greater financial flexibility, enabling them to absorb unexpected financial shocks while maintaining scheduled loan repayments.

The dashboard enables executives to evaluate how applicant income influences lending outcomes across different customer segments.

Income analysis assists financial institutions in:

- Determining affordable loan amounts
- Reducing repayment stress
- Improving portfolio quality
- Enhancing responsible lending practices

By aligning loan approvals with demonstrated repayment capacity, institutions strengthen both commercial performance and regulatory compliance.

Work Experience Analysis

Years of professional experience often provide additional evidence regarding borrower financial stability.

Applicants with longer employment histories generally demonstrate:

- Career stability
- Consistent income generation
- Lower employment uncertainty
- Improved repayment capability

The dashboard evaluates whether work experience contributes meaningfully to loan approval decisions.

These insights assist management in refining borrower assessment models while improving future lending decisions through more comprehensive risk evaluation.

Demographic Analysis

Understanding borrower demographics enables financial institutions to evaluate lending patterns across different customer groups while ensuring lending practices remain fair, objective, and non-discriminatory.

The dashboard analyses applicant characteristics including:

- Age
- Gender
- Educational qualification

These dimensions provide additional context regarding borrower profiles but should complement—not replace—financial indicators such as income, employment, and credit history.

Age Analysis

Age distribution enables management to evaluate lending demand across different life stages.

Younger applicants may demonstrate strong future earning potential but relatively limited financial history.

Older applicants often possess longer employment histories and stronger financial records but may have different borrowing objectives.

Understanding these demographic patterns assists product development, customer segmentation, and long-term lending strategy.

Gender Analysis

Gender analysis enables management to monitor portfolio diversity while ensuring lending decisions remain unbiased and compliant with regulatory standards.

The analysis should not be interpreted as evidence that gender influences creditworthiness.

Instead, it provides transparency regarding customer composition and supports responsible lending practices.

Objective lending decisions should always prioritise measurable financial indicators over demographic characteristics.

Educational Qualification Analysis

Educational attainment may provide indirect insight into future income potential, career progression, and financial stability.

The dashboard enables executives to examine whether educational qualifications are associated with observable differences in lending outcomes.

Although education alone does not determine repayment behaviour, combining educational information with employment history and income creates a more comprehensive borrower profile.

Executive Interpretation

Collectively, the dashboard demonstrates that successful lending depends upon evaluating multiple borrower characteristics simultaneously rather than relying on a single approval criterion.

The analysis indicates that responsible lending decisions emerge from balancing:

- Creditworthiness
- Employment stability
- Income capacity
- Financial behaviour
- Demographic context
- Portfolio diversification

This holistic approach strengthens portfolio quality while supporting sustainable business growth.

From an executive perspective, the dashboard serves as a strategic decision-support tool that enables management to monitor lending performance, identify emerging risks, optimise credit policies, and improve long-term financial performance through evidence-based credit risk management.

Credit Risk Assessment

Credit risk assessment forms the foundation of prudent lending and represents one of the most critical responsibilities of financial institutions. Every approved loan carries an inherent level of uncertainty because future repayment behaviour cannot be guaranteed. Consequently, organisations must continuously evaluate borrower characteristics to balance commercial growth with acceptable levels of financial risk.

The dashboard demonstrates how Business Intelligence can transform thousands of historical loan applications into meaningful credit risk insights that support consistent, objective, and evidence-based lending decisions.

Rather than relying solely on manual judgement, executives can evaluate multiple borrower characteristics simultaneously to determine the likelihood of successful loan repayment while protecting the institution from excessive credit losses.

The analysis considers several key credit risk indicators, including:

- Credit Score
- Employment Status
- Annual Income
- Years of Work Experience
- Educational Qualification

- Age Profile
- Geographic Distribution
- Loan Approval History

Collectively, these variables provide a comprehensive assessment of borrower quality and enable management to strengthen lending decisions while maintaining portfolio stability.

Creditworthiness Assessment

Creditworthiness represents the borrower's overall ability and willingness to repay financial obligations.

The dashboard indicates that approved applicants generally possess stronger financial profiles characterised by:

- Better credit history
- Stable employment
- Sustainable income
- Greater work experience
- Lower perceived default risk

These characteristics significantly improve repayment probability and reduce expected credit losses throughout the loan lifecycle.

Conversely, applicants demonstrating weaker financial indicators present increased lending uncertainty and therefore experience higher rejection rates under the institution's credit approval policy.

This disciplined assessment process supports responsible lending while protecting both depositors and shareholders from avoidable financial losses.

Portfolio Quality Assessment

The quality of a loan portfolio depends not on the quantity of loans approved but on the long-term repayment performance of borrowers.

Although approving a larger number of applications may increase short-term lending revenue, excessive approval of high-risk borrowers often leads to:

- Higher default rates
- Increased loan impairment
- Greater provisioning requirements
- Reduced profitability
- Capital erosion

The dashboard suggests that the institution maintains a conservative lending strategy designed to prioritise portfolio quality over aggressive loan growth.

Such an approach promotes sustainable profitability while strengthening long-term financial stability.

Risk Segmentation

One of the principal advantages of Business Intelligence is the ability to segment borrowers into meaningful risk categories.

Rather than treating every applicant equally, financial institutions can classify customers according to their estimated repayment capability.

Typical borrower segments include:

Low-Risk Borrowers

Characteristics generally include:

- Strong credit scores
- Stable employment
- Consistent income
- Extensive work experience
- Positive financial history

These customers typically qualify for faster loan approvals, competitive interest rates, and larger borrowing limits.

Moderate-Risk Borrowers

Moderate-risk applicants may possess acceptable financial characteristics but require additional assessment before approval.

Examples include:

- Limited credit history
- Moderate income
- Short employment duration
- Average credit scores

Additional documentation or guarantors may reduce lending uncertainty.

High-Risk Borrowers

Applicants demonstrating multiple financial risk indicators require greater lending caution.

Risk characteristics may include:

- Poor credit scores
- Unstable employment
- Limited income
- Short work experience
- Previous repayment challenges

These borrowers present elevated default probability and therefore require stricter credit assessment procedures.

Lending Policy Evaluation

The dashboard provides management with valuable insight into the effectiveness of current lending policies.

The relatively selective approval rate indicates that existing credit policies successfully identify lower-risk borrowers while limiting exposure to higher-risk applicants.

However, lending policies should remain dynamic rather than static.

Changes in economic conditions, employment trends, inflation, and borrower behaviour require continuous monitoring and periodic policy review.

Regular analytical evaluation ensures lending criteria remain commercially competitive without compromising acceptable levels of portfolio risk.

Predictive Credit Analytics

Historical lending data provides valuable opportunities for future predictive modelling.

While this project focuses primarily on descriptive and diagnostic analytics, the same dataset could support predictive models capable of estimating future default probability before lending decisions are made.

Potential predictive techniques include:

- Logistic Regression
- Decision Trees
- Random Forest
- Gradient Boosting
- Machine Learning Credit Scoring Models

Predictive analytics enables institutions to identify high-risk borrowers earlier while improving lending consistency and reducing future impairment losses.

This represents a natural progression from traditional reporting towards Artificial Intelligence-driven credit decision support.

Geographic Analysis

Geographic analysis evaluates lending activity across different cities and locations to determine how loan demand varies geographically.

Understanding geographic lending patterns enables financial institutions to optimise branch operations, improve regional marketing strategies, diversify portfolio exposure, and identify emerging commercial opportunities.

The dashboard illustrates the distribution of loan applications across multiple cities, providing executives with a clearer understanding of market penetration and regional lending demand.

Geographic Lending Demand

Different geographic locations often demonstrate varying borrowing behaviours due to differences in:

- Population density
- Employment opportunities
- Household income
- Economic development
- Business activity
- Cost of living

Monitoring regional lending demand enables management to allocate organisational resources more effectively while supporting balanced portfolio growth.

Cities demonstrating consistently high lending activity may justify:

- Additional branch investment
- Expanded lending capacity
- Increased relationship management
- Enhanced customer acquisition programmes

Meanwhile, lower-demand regions may require targeted marketing initiatives or revised product offerings.

Geographic Portfolio Diversification

Concentration risk occurs when excessive lending exposure exists within a limited number of geographic markets.

Economic downturns affecting a particular region may significantly increase default rates if the institution's portfolio lacks sufficient geographic diversification.

The dashboard enables executives to monitor geographic concentration by identifying regions responsible for the largest proportion of loan applications.

Maintaining a geographically diversified loan portfolio reduces exposure to local economic shocks while improving long-term portfolio resilience.

Regional Business Opportunities

Geographic analytics extends beyond risk management by identifying future commercial opportunities.

Management can utilise lending demand patterns to:

- Expand branch networks

- Introduce regional lending products
- Improve customer acquisition
- Strengthen local partnerships
- Increase financial inclusion

These insights enable financial institutions to align commercial expansion strategies with actual customer demand rather than assumptions.

Executive Interpretation

Geographic analysis demonstrates that lending demand is not evenly distributed across markets.

By continuously monitoring regional lending activity, executives can optimise resource allocation, improve portfolio diversification, reduce concentration risk, and strengthen long-term business growth.

Geographic Business Intelligence therefore supports both commercial expansion and prudent portfolio management.

Business Impact

The analytical insights generated through this project extend well beyond operational reporting and provide measurable strategic value across multiple business functions.

Credit risk analytics enables financial institutions to replace subjective lending decisions with objective, evidence-based assessments that improve consistency, efficiency, and financial performance.

Several important business observations emerge from this analysis.

- Loan approvals remain deliberately selective, supporting portfolio quality.
- Credit scores play a central role in borrower assessment.
- Employment stability significantly influences lending outcomes.
- Income and work experience strengthen borrower evaluation.
- Geographic analysis identifies opportunities for market expansion.
- Business Intelligence improves lending transparency and executive oversight.

Collectively, these findings strengthen organisational decision-making while supporting responsible lending practices.

Operational Impact

The dashboard enables lending teams to process applications more consistently by providing objective borrower assessment criteria.

Operational benefits include:

- Faster lending decisions
- Improved underwriting consistency
- Reduced manual assessment

- Enhanced reporting efficiency
- Better portfolio monitoring
- Stronger regulatory documentation

These improvements increase organisational productivity while reducing operational risk.

Strategic Impact

From a strategic perspective, the project enables executives to align lending operations with broader organisational objectives.

Business Intelligence supports:

- Sustainable portfolio growth
- Improved customer segmentation
- Risk-adjusted lending
- Better capital allocation
- Stronger executive decision-making
- Enhanced regulatory compliance

Rather than reacting to loan defaults after they occur, executives can proactively identify emerging risks before significant financial losses develop.

Customer Impact

Data-driven lending decisions also improve customer experience.

Objective assessment criteria reduce inconsistencies while enabling faster loan processing and greater transparency throughout the approval process.

Improved analytical capability supports:

- Fair lending practices
- Consistent approval decisions
- Better customer communication
- Reduced processing time
- Increased customer confidence

Responsible lending ultimately benefits both financial institutions and borrowers by encouraging sustainable borrowing behaviour.

Financial & Accounting Perspective

One of the distinguishing strengths of this project is its integration of Business Intelligence with accounting and financial management principles.

Credit risk analysis directly influences an institution's financial performance because lending decisions affect revenue generation, impairment costs, liquidity, regulatory capital, and shareholder value.

Understanding these financial relationships enables executives to evaluate lending performance from both operational and financial perspectives.

Impact on the Income Statement

Approved loans generate interest income, which represents one of the primary revenue sources for financial institutions.

However, profitability depends not only on loan growth but also on borrower repayment performance.

Effective credit assessment contributes to:

- Higher Interest Revenue
- Lower Bad Debt Expense
- Reduced Loan Loss Provisions
- Improved Operating Profit
- Increased Net Profit

Conversely, approving excessive numbers of high-risk borrowers increases impairment expenses, reducing overall profitability.

Strong credit analytics therefore protects earnings while supporting sustainable revenue generation.

Impact on the Balance Sheet

Credit risk management influences several major Balance Sheet accounts.

Loan Assets

Approved loans become financial assets recorded on the Balance Sheet.

High-quality loan portfolios strengthen asset quality while improving long-term financial stability.

Allowance for Credit Losses

Financial institutions establish provisions for expected loan losses in accordance with accounting standards such as IFRS 9.

Improved borrower assessment reduces expected credit losses and lowers impairment allowances.

This strengthens both asset quality and shareholder confidence.

Cash and Liquidity

Successful loan repayments generate cash inflows that improve organisational liquidity.

Healthy repayment performance supports:

- Stronger cash balances

- Improved liquidity ratios
- Increased lending capacity
- Greater financial flexibility

Shareholders' Equity

Lower impairment expenses and stronger profitability increase retained earnings.

Improved retained earnings strengthen shareholders' equity while supporting future business expansion.

Impact on the Cash Flow Statement

Loan repayments generate positive operating cash inflows throughout the lending cycle.

Strong portfolio performance improves:

- Operating Cash Flow
- Liquidity Management
- Capital Availability
- Investment Capacity

Conversely, higher default rates delay or eliminate expected cash inflows, creating liquidity pressure despite reported accounting profits.

This demonstrates why credit quality remains as important as loan growth.

Regulatory Perspective

Modern financial institutions operate within strict regulatory frameworks designed to protect depositors, investors, and financial system stability.

Effective credit risk analytics supports compliance with:

- IFRS 9 Expected Credit Loss (ECL) principles
- Basel capital adequacy requirements
- Internal risk governance policies
- Responsible lending regulations
- Audit and regulatory reporting requirements

Business Intelligence therefore contributes not only to commercial performance but also to sound corporate governance and regulatory compliance.

Business Intelligence Value

This project demonstrates how Business Intelligence transforms traditional credit reporting into executive decision support.

Rather than answering only:

"How many loans were approved?"

the dashboard also addresses:

- Why were certain applicants approved?
- Which borrower characteristics reduce default risk?
- How strong is the overall loan portfolio?
- What financial risks exist?
- Which markets offer future lending opportunities?
- How do lending decisions affect profitability and capital?

This progression from descriptive reporting to strategic credit risk management reflects the evolving role of Data Analysts within banking, fintech, consulting, and multinational organisations.

The project demonstrates the ability to combine SQL, Power BI, financial analysis, business intelligence, accounting knowledge, and executive storytelling into a comprehensive analytical solution capable of supporting high-quality lending decisions and sustainable organisational growth.

Strategic Recommendations

The findings from this analysis demonstrate that effective credit risk management extends beyond approving or rejecting loan applications. Sustainable lending requires balancing commercial growth with prudent risk management, regulatory compliance, and long-term financial stability.

Based on the insights generated from the dashboard, the following strategic recommendations are proposed.

1. Adopt a Risk-Based Lending Framework

Rather than applying uniform lending criteria to all applicants, the institution should implement a risk-based lending framework that classifies borrowers according to their estimated probability of default.

Potential borrower categories include:

- Low Risk
- Moderate Risk
- High Risk

This approach enables the organisation to:

- Improve lending precision
- Optimise interest pricing
- Reduce unnecessary loan rejections
- Maintain portfolio quality
- Increase profitability through risk-adjusted lending

Risk segmentation also supports more personalised lending products tailored to different customer profiles.

2. Enhance Credit Scoring Models

The analysis confirms that credit score remains one of the strongest indicators of repayment behaviour.

Management should continue investing in advanced credit scoring models by incorporating additional behavioural variables such as:

- Existing debt obligations
- Debt-to-income ratio
- Repayment history
- Banking transaction behaviour
- Savings patterns
- Loan utilisation history

Integrating multiple financial indicators improves predictive accuracy while reducing future loan defaults.

3. Introduce Predictive Credit Risk Analytics

Current reporting primarily explains historical lending performance.

Future analytical capabilities should incorporate predictive analytics capable of estimating borrower default probability before loan approval.

Suitable analytical models include:

- Logistic Regression
- Decision Trees
- Random Forest
- Gradient Boosting
- Machine Learning Classification Models

Predictive modelling enables lending institutions to move from reactive reporting towards proactive credit risk management.

4. Continuously Monitor Portfolio Performance

Credit portfolios evolve continuously as economic conditions change.

Management should establish regular dashboard monitoring to evaluate:

- Approval trends
- Portfolio quality
- Geographic concentration
- Customer risk migration
- Default indicators
- Lending policy effectiveness

Continuous monitoring enables early identification of emerging risks before they materially affect financial performance.

5. Improve Geographic Portfolio Diversification

Geographic analysis indicates that lending demand varies across different regions.

Management should seek to balance portfolio exposure by avoiding excessive concentration within a limited number of markets.

A geographically diversified portfolio reduces exposure to local economic downturns while strengthening long-term resilience.

6. Expand Responsible Lending Practices

Responsible lending benefits both financial institutions and borrowers.

Rather than focusing solely on loan approvals, management should encourage lending practices that promote sustainable repayment behaviour.

Examples include:

- Financial literacy programmes
- Customer affordability assessments
- Appropriate loan sizing
- Improved borrower education
- Transparent lending communication

Responsible lending reduces future defaults while strengthening customer relationships.

7. Strengthen Executive Decision Support

Business Intelligence dashboards should become a routine component of executive reporting.

Interactive dashboards enable leadership teams to:

- Monitor portfolio quality
- Evaluate lending performance
- Compare regional trends
- Assess financial exposure
- Support strategic planning

Replacing static reports with interactive analytics improves organisational agility and evidence-based decision-making.

8. Integrate Credit Analytics with Enterprise Data

Future development should integrate lending data with additional enterprise datasets, including:

- Customer relationship management (CRM)

- Transaction history
- Digital banking activity
- Fraud monitoring systems
- Treasury data
- Collections performance

Enterprise-wide analytics provides a more comprehensive understanding of customer behaviour and supports stronger lending decisions.

Skills Demonstrated

This project demonstrates a combination of technical expertise, business analysis, financial knowledge, and executive communication skills expected of modern Data Analysts, Business Intelligence Analysts, Financial Analysts, and Business Analysts.

Data Analytics

- Data Cleaning
- Data Transformation
- Exploratory Data Analysis (EDA)
- Data Validation
- Business Intelligence
- Statistical Interpretation
- Executive Reporting

SQL

- Data Extraction
- Data Filtering
- Aggregation
- Joins
- Grouping
- Sorting
- Query Optimisation
- Relational Database Analysis

Power BI

- Interactive Dashboard Development
- Data Modelling
- DAX Calculations
- KPI Design
- Executive Visualisation
- Report Navigation
- Business Storytelling

Credit Risk Analytics

- Loan Portfolio Analysis
- Credit Risk Assessment

- Borrower Segmentation
- Loan Approval Analysis
- Risk Identification
- Lending Performance Evaluation
- Portfolio Quality Assessment

Financial Analysis

- Financial Performance Evaluation
- Revenue Interpretation
- Portfolio Profitability
- Credit Loss Assessment
- Business Performance Measurement
- Financial Decision Support

Business Analysis

- Requirement Interpretation
- Business Problem Solving
- Strategic Recommendations
- Executive Decision Support
- Process Evaluation
- Stakeholder Communication

Data Storytelling

One of the primary objectives of this project was not simply to analyse data but to communicate business insights in a manner that supports executive decision-making.

The dashboard combines analytical findings with business interpretation, transforming technical outputs into actionable recommendations for senior management.

This ability to bridge the gap between technical analytics and business strategy is increasingly valuable within banking, consulting, fintech, insurance, healthcare, and multinational organisations.

Why This Project Matters

Modern organisations generate vast quantities of operational and financial data every day.

However, data alone creates little business value unless it is transformed into meaningful information capable of supporting strategic decisions.

This project demonstrates how Business Intelligence enables organisations to:

- Improve lending decisions
- Reduce financial risk
- Strengthen portfolio quality
- Enhance profitability
- Support regulatory compliance
- Improve executive reporting

- Enable evidence-based decision-making

Rather than functioning solely as a technical dashboard, this solution illustrates how analytics contributes directly to commercial success and organisational sustainability.

The project also demonstrates the importance of combining analytical thinking with financial knowledge and business understanding.

By integrating SQL, Power BI, accounting principles, and business intelligence, the analysis provides executives with a comprehensive view of lending performance that supports long-term strategic planning.

This reflects the evolving role of modern Data Analysts, who are increasingly expected to influence business outcomes rather than simply produce reports.

About the Author

Dayo Rex Afariogun

Senior Business Analyst | Financial Data Analytics & Business Intelligence

Dayo Rex Afariogun is a Business Intelligence and Data Analytics professional with over 14 years of experience supporting business growth through data-driven decision-making, financial analysis, operational improvement, and strategic business reporting.

With an academic background in **Accounting** and professional expertise in **SQL, Power BI, Python, Microsoft Excel, and Business Intelligence**, he specialises in transforming complex datasets into executive-level insights that improve organisational performance.

His work focuses on designing analytical solutions that bridge the gap between technical data analysis and business strategy across industries including:

- Financial Services
- Banking
- Credit Risk
- Commercial Analytics
- Healthcare
- Sales & Revenue Analytics
- Business Consulting

His portfolio demonstrates practical experience in:

- Credit Risk Analytics
- Loan Portfolio Analysis
- Commercial Revenue Intelligence
- Healthcare Cost Analysis
- Financial Reporting
- Sales Performance Analysis
- SQL Database Analysis
- Interactive Dashboard Development

- Executive Decision Support

Dayo is passionate about leveraging analytics to solve complex business challenges and is actively seeking opportunities to contribute to forward-thinking organisations where data-driven decision-making creates measurable business value.

Connect with me:

- **LinkedIn:** www.linkedin.com/in/dayo-afariogun

- **GitHub:** github.com/dayo-data-analyst

Executive Conclusion

This project demonstrates how Business Intelligence can transform historical lending data into actionable executive insight that supports responsible credit risk management and sustainable financial growth.

Using **Microsoft SQL Server** for data extraction and **Power BI** for dashboard development and visualisation, the analysis examined **5,000 loan applications** to identify the factors influencing lending decisions, borrower risk, and overall portfolio quality. The findings show that the institution maintains a disciplined lending strategy, approving approximately **23%*** of applications while rejecting **77%**, reflecting a strong emphasis on prudent underwriting and long-term portfolio stability.

Beyond descriptive reporting, the project illustrates how combining borrower characteristics—including credit score, employment status, income, work experience, demographics, and geographic distribution—provides a holistic understanding of credit risk. These insights enable executives to optimise lending policies, improve capital allocation, strengthen regulatory compliance, and reduce expected credit losses while supporting sustainable profitability.

From a technical perspective, the project demonstrates proficiency in SQL, Power BI, DAX, data modelling, dashboard design, and business intelligence reporting.

From a business perspective, it highlights the ability to translate analytical findings into strategic recommendations that improve operational efficiency and executive decision-making.

Ultimately, this project reflects the evolving role of modern Data Analysts—not only as technical specialists, but as strategic business partners who transform data into measurable organisational value.

Key Technologies

- Microsoft SQL Server
- SQL
- Power BI
- Power Query
- DAX
- Microsoft Excel
- Business Intelligence
- Data Analytics

Core Competencies Demonstrated

- Credit Risk Assessment
- Loan Portfolio Analysis
- Business Intelligence
- Financial Analytics
- Executive Reporting
- Data Visualisation
- Dashboard Development
- Data Storytelling
- SQL Querying
- Data Modelling
- KPI Design
- Business Analysis
- Decision Support
- Strategic Thinking

Thank you for taking the time to review this project.

I welcome opportunities to discuss how my experience in **data analytics, business intelligence, financial analysis, and executive reporting** can help organisations make smarter, data-driven decisions and deliver measurable business outcomes.